

Aya Almalla

Artificial Intelligence Engineer

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PROFESSIONAL SUMMARY

Building AI solutions that solve real-world challenges by turning ideas into practical, deployable products. Artificial Intelligence Engineering graduate with hands-on experience in computer vision, NLP, RAG systems, and end-to-end AI development across different domains, from dataset curation and model selection to evaluation, deployment, and product integration. Experienced in leading technical teams and connecting AI development with product and market needs.

EDUCATION

Damascus University - Bachelor of Science in Information Technology Engineering, Artificial Intelligence Specialization
Sep 2021 - Aug 2026

TECHNICAL SKILLS

Programming: Python, Dart

AI & Machine Learning: Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, RAG, Image Classification, Object Detection, Segmentation, Dataset Curation, Data Preprocessing, Fine-Tuning, Model Evaluation, Error Analysis

Frameworks & Libraries: PyTorch, TensorFlow, Hugging Face Transformers, OpenCV, YOLO, U-Net++, ResNet34, Sentence-Transformers, NumPy, Pandas

AI Engineering & Deployment: FastAPI, REST APIs, Inference Pipelines, Model Deployment, Modal, Docker, Streamlit

Data & Software Tools: ChromaDB, PostgreSQL, Kafka, Firebase, Git, GitHub, Flutter, BLoC/Cubit

Professional Skills: Problem Solving, Analytical Thinking, Research, Communication, Teamwork, Leadership, Collaboration, Product Thinking

APPLIED EXPERIENCE

AI / Computer Vision Trainee & Team Leader - EyeAI, MozaicAI, Remote **Apr 2026 - Aug 2026**

- Led the project team and owned the AI workstream for an AI-assisted retinal screening and clinical decision-support platform; the project was recognized as Best Team, and I received the Best Leader award.
- Curated and cleaned retinal datasets, selected suitable model architectures, and evaluated YOLO26-based classification/detection/segmentation approaches alongside U-Net++ with a ResNet34 encoder.
- Achieved a 0.93 Macro AUC for five-grade diabetic retinopathy classification.
- Produced a comprehensive Syrian market and go-to-market study covering target segments, competitor and SWOT analysis, positioning, business model and pricing, partnerships, risk mitigation, and KPIs.

Contract Computer Vision Engineer - Medical AI, AI-Khatib Lab **Mar 2025 - Sep 2025**

- Developed computer vision components for PlateMate, an AI-powered application automating Antibiotic Susceptibility Testing for a real laboratory client, including disc detection, inhibition-zone segmentation, automated diameter measurement, and reporting.
- Achieved 0.90+ precision for disc detection, 0.94 Dice, and 0.91 IoU for segmentation.

PROJECTS

CADify: AI-Powered Floor Plan to CAD & 3D Conversion

Tech: Python, PyTorch, Computer Vision, Segmentation, OpenCV, FastAPI, Modal

- Developed an AI-powered system that converts raster architectural floor plans into structured, editable DXF/CAD files compatible with AutoCAD workflows, with support for 3D scene generation.
- Owned and developed the complete topology reconstruction pipeline, transforming model outputs into connected walls, junctions, openings, rooms, and structured CAD/3D handoff data.
- Achieved 0.93 F1 for wall reconstruction and 0.99 F1 / 0.96 IoU for door and window reconstruction.
- Integrated the pipeline through FastAPI and deployed model inference workloads on Modal.

Shifaa: Arabic Medical Question Classification

Tech: Python, PyTorch, Hugging Face, AraBERT, Streamlit

- Built an end-to-end Arabic NLP pipeline for 16 medical categories, performing EDA, class-distribution and text analysis, data-leakage checks, and benchmarking TF-IDF/traditional ML, CNN/LSTM, and AraBERT approaches.
- AraBERT achieved 0.82 accuracy and 0.70 macro-F1; deployed the final model on Hugging Face and integrated it into an interactive Streamlit app with confidence scores.

Lecture-Saver 3000: RAG PDF Assistant

Tech: Python, ChromaDB, Sentence-Transformers, Gemini API, Streamlit

- Built a RAG assistant for course PDFs using vector retrieval and an LLM, implementing document ingestion, chunking, metadata-based retrieval, and file/page-level source citations.

Equipment Utilization & Activity Classification Prototype

Tech: Python, YOLOv8, OpenCV, Kafka, PostgreSQL, Streamlit, Docker

- Built a video-based system for tracking excavator activity and measuring equipment utilization, with dashboard analytics for active time, idle time, utilization rate, and machine activity states.

ADDITIONAL PROJECTS

- Automated Zuma Player | Python, OpenCV: Real-time classical computer vision automation using optical flow, circle detection, color classification, and target selection.
- Antibiotic Expert System AI | Python, Experta: Rule-based medical decision-support prototype for antibiotic recommendation and dosage calculation.
- Voyago | Flutter, Firebase, Stripe, Google Maps API: Travel mobile application with trip customization, booking, payments, maps, and real-time travel utilities.

COURSES & PROFESSIONAL DEVELOPMENT

- Aga Khan Foundation - Youth Entrepreneurship and Employable Skills (YES)** Aug 2026
 - Training in entrepreneurship and product development, including Business Model Canvas (BMC), market understanding, opportunity validation, and turning ideas into executable business concepts.
- Global Entrepreneurship Week Syria - Startup Syria, Entrepreneurship Bootcamp** Jan 2026
 - Completed a 3-day intensive program focused on startup idea development, market positioning, and marketing strategy.
- Machine Learning by Andrew Ng - Coursera** 2024
- Dart & Mobile Development Practices** 2023 - 2024
 - Freezed modeling, JSON serialization, lazy loading, caching, network error handling, and localization for scalable Flutter applications.

LEADERSHIP & VOLUNTEERING

- HR Manager - RBCs Team** Sep 2022 - Mar 2023
 - Managed volunteer recruitment, onboarding, training, and team support for freshman members.
- Content Writer - PTC at Damascus University** Sep 2023 - 2024
 - Created educational and technical content to support student learning and community engagement.
- Organizing Team Member - 2024 Projects Exhibition** 2024
 - Assisted with event organization and coordination for the faculty projects exhibition.

LANGUAGES

Arabic: Native | English: Professional Working Proficiency